

## RESEARCH ARTICLE

# A Compact Piezo-Drive Rotatable Scanning Tunneling Microscope in a 12 T Cryogen-Free Magnet

Junwei Liu<sup>1</sup>  | Zihao Li<sup>1</sup> | Dan Wu<sup>1,2</sup> | Shaofeng Zheng<sup>1</sup> | Yue Gao<sup>1</sup> | Shiwei Yang<sup>1</sup> | Jihao Wang<sup>2,3</sup> | Jing Zhang<sup>2,3</sup> | Qiyuan Feng<sup>2,3</sup> | Yubin Hou<sup>2,3</sup>  | Wenjie Meng<sup>2,3</sup> | Yalin Lu<sup>1,4</sup> | Qingyou Lu<sup>1,2,3,4</sup>

<sup>1</sup>University of Science and Technology of China, Hefei, People's Republic of China | <sup>2</sup>Anhui Key Laboratory of Low-energy Quantum Materials and Devices, High Magnetic Field Laboratory, HFIPS, Chinese Academy of Sciences, Hefei, People's Republic of China | <sup>3</sup>The High Magnetic Field Laboratory of Anhui Province, Hefei, People's Republic of China | <sup>4</sup>Anhui Laboratory of Advanced Photon Science and Technology and Hefei National Laboratory, University of Science and Technology of China, Hefei, People's Republic of China

**Correspondence:** Wenjie Meng ([wjmeng@hmfll.ac.cn](mailto:wjmeng@hmfll.ac.cn)) | Qingyou Lu ([qxl@ustc.edu.cn](mailto:qxl@ustc.edu.cn))

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## ABSTRACT

Atomically resolved scanning tunneling microscope (STM) capable of in situ rotation in a narrow magnet bore has become a long-awaited but challenging technique in the field of strong correlation studies since it can introduce the orientation of the strong magnetic field as a control parameter. This article presents the design and functionality of a piezoelectrically driven rotatable STM (RSTM), operating within a 12 T cryogen-free magnet and achieving a base temperature below 1.8 K, along with spectroscopic capabilities. The system features a compact STM head unit that combines an inertia drive shaft with spring clamping onto the inner wall of a slender piezoelectric scanning tube (PST), enabling both stepper and scanner functionality while reducing the STM's size to 25.5 mm in length and 9 mm in diameter, facilitating rotation within the magnet bore. Another linear piezoelectric motor, driven by a PST, employs a mechanical linkage to convert linear into rotational motion, driving the STM head unit coaxially aligned with it. This mechanism enables STM accurate rotation, offering angle control from 0° to 90° with an ideal closed-loop accuracy of 0.11° per 0.01 V, as determined by a calibrated Hall sensor. Compact and suspended as a standalone unit at the tail of the sample probe, the RSTM is effectively shielded from external mechanical vibrations via secondary counterweight damping. To validate the device performance, the topographic images of graphite and NbSe<sub>2</sub> and their spectroscopy at various magnetic field orientations up to 12 T and temperatures below 1.8 K are obtained. The compact and vibration-resistant RSTM provides compatibility with ultra-high-field water-cooled magnets, facilitating investigations into multiangle magnetic field modulation studies for condensed matter physics.

## 1 | Introduction

The scanning tunneling microscope (STM) is an advanced technology, which is promising to the exploration of the quantum materials nowadays as the topological insulators (Jiao et al. 2018; Pirie

et al. 2020), superconductors (Edkins et al. 2019; Song et al. 2011), and transition metal dichalcogenides (Tilak et al. 2023; Zhou et al. 2016), advancing the researches of the localized properties including the atomic resolution of the small area, the analysis of the band structure around the defects, and the edge states around the

## Summary

- The compact structure, which can be applied at the narrow-bore magnet, has the capacity to provide a higher magnetic field, promising to the researches for the materials in harsher environments at low temperatures.
- The rotating unit is combined with a piezo-controlling part, which has the high accuracy to control the rotating angle compared with the mechanical way to control the rotation, exhibiting powerful controlling of angle, and the angle can be reached by measuring the output of the Hall element.
- The maximum magnetic field can be applied up to 12 T at the in-plane orientation, which is critical to numerous materials including the modulation of the superconductor based on its high in-plane critical field.

boundary. This technology has been developed with a long history, and there are plenty intriguing variations: the spin-polarized STM (Li et al. 2023; Wortmann et al. 2001), which enables to recognize the polarized tunneling on the surface, the Josephson STM (Du et al. 2020; Liu et al. 2021a), which maintains the high resolution and allows the visualization of the tunneling of Cooper pair, and the CO-tip-functionalized STM (Liu et al. 2022, 2019), which enhances the resolution of the organic molecular and the elaborate bonds of them. Moreover, the quasi-particle interference (Haenke et al. 2012; Petersen et al. 1998), the mapping of the local density of states (Pan et al. 2000; Stolyarov et al. 2018), and the inelastic tunneling spectrum (Hahn, Lee, and Ho 2000; She et al. 2013) also expand the capabilities to uncover fruitful electronic states around the confined region. With the development of the realization of many overall properties of quantum materials, STM nowadays can still reveal the detailed information, which cannot be directly observed or submerge in the materials such as the pair density wave (Gu et al. 2023; Liu et al. 2021b), the nematic phase (Nie et al. 2022; Rubio-Verdu et al. 2022), and the spin selected tunneling (Aishwarya et al. 2022; Takahashi et al. 2016). The advantages of STM are based on two significant external conditions: the low temperature and the high magnetic field. The low temperature is very crucial to copious novel phenomena because numerous intrinsic physical characteristics only emerge when the thermal fluctuation is mild enough, inclusive of the formation of the Landau level (LL) with the condition  $k_B T < \hbar \omega$  (Gogoi et al. 2017), the creation of the pairing between electrons under the critical temperature  $T_C$  in superconductors (Tissen et al. 2013), and the establishment of the commensurate charge density wave beneath the temperature  $T_{Com}$  (Naito and Tanaka 1982). The high magnetic field permits another dimension to modulate the attributes of the electronic states in the system, triggering the quantum Hall effect within the topological insulators (Xu et al. 2014; Yoshimi et al. 2015), the vortex in the type-II superconductors (Menghini, Fasano, and de la Cruz 2002; Pasquini et al. 2008), and the Fulde-Ferrell-Larkin-Ovchinnikov (FFLO) states in the unconventional superconductors (Kumagai et al. 2006; Uji et al. 2006).

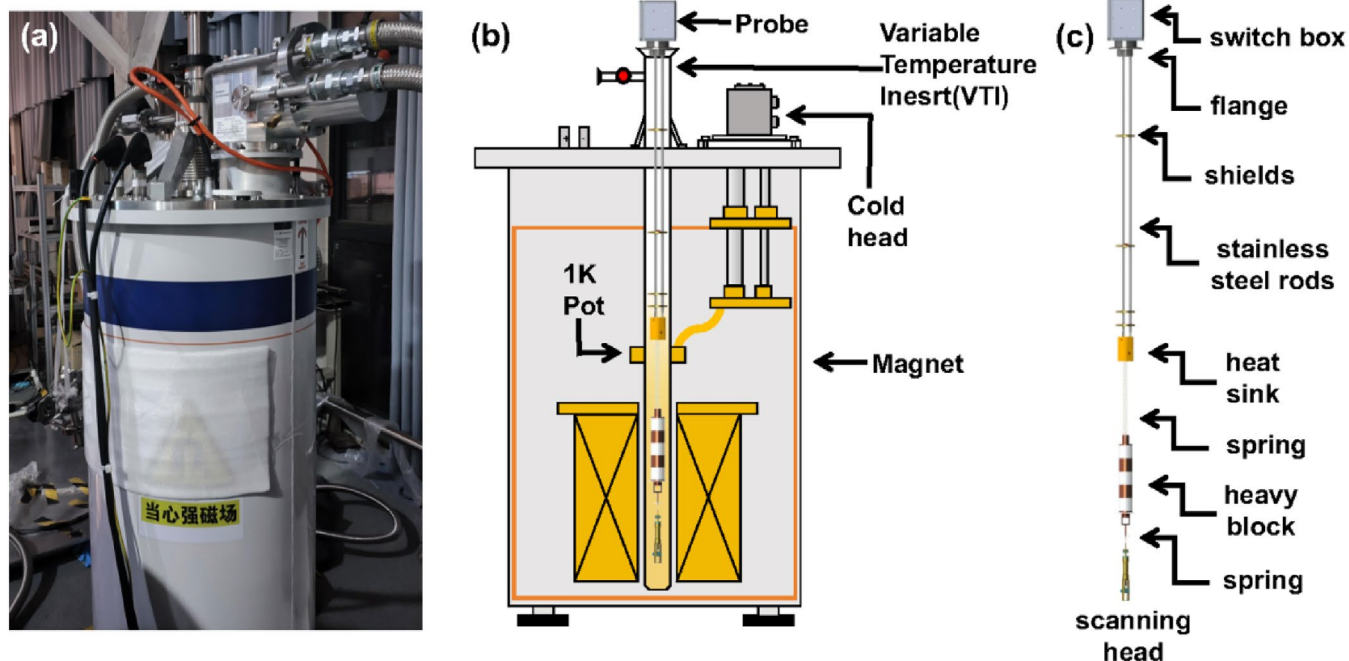
The STM equipped with the magnet and low temperature dewar has been established for the long history, and the field has reached 30 T for the maximum (Meng et al. 2020) and the

base temperature can also be 20 mK for advanced study (Misra et al. 2013), but the orientation of the field can only be fixed at the vertical direction, which largely limits the observation of many localized in-plane modulation of the materials, remaining the abundant opportunities to discover new physics. Although the vector magnet (Galvis et al. 2015; Zhou et al. 2023) can ease the problem by the rotatable field at any directions, it is still discouraged by the amplitude, which is limited to 5 T at maximum (Trainer et al. 2017), to stimulate some unusual phenomena. Besides, the STM nowadays is often applied to the superconductor magnet, which has the advantages of subdued mechanical and electronic noise but with large consumption of the liquid helium, being too lavish to implement the scientific experiments.

This article reports the in situ rotatable STM (RSTM) that can rotate from  $0^\circ$  to  $90^\circ$  at the dry magnet whose field is 12 T at maximum at low temperature minimum to 1.63 K by the driving of “spider driver,” which is assembled by the 52 mm piezo tube, the zicronia tube, the Beryllium copper spring, and the tantalum shaft. Compelled by the appropriate voltage from the controller, the rotation can reach the accuracy of  $0.16^\circ$  per 0.01 V (the maximum resolution of the voltmeter) at 4 T and 1.73 K with the activation of 4.7 mA and 6 V for the Hall element. Connected to the spring-connected heavy block with another spring for the multistage diminution of the mechanical noises, the scanning head can remain stable for the topography and spectrum even under the harsh circumstance induced by the cold head on the magnet, laying the solid foundation for implementation of STM in the cryogen-free magnet in the future. With its tiny structure, the RSTM can be applied to the restricted space, assuring the future application to the water-cool magnet and the hybrid magnet. As for the performance of the instrument, this article has measured the graphite for its topography and the spectrum for in-plane, the out-of-plane as well as the 4 T-rotary field at 1.73 K and the topography of bulk NbSe<sub>2</sub> for its topography and the spectrum for in-plane magnetic field at 1.63 K. From the result, this article has found the obvious difference from the spectrum of graphite and find the preserve of the gap of the NbSe<sub>2</sub> under its in-plane critical field. With the capabilities exhibited above, the device is stable enough to be applied to other researches and can modulate more intriguing properties of the materials. Besides, the rotating unit can also carry the magnetic force microscope (MFM), which plays a significant role in the investigation of the magnetic Van der Waals materials (Deng et al. 2023; Zhu et al. 2022), expanding more functionalities of the angle-rotating design.

## 2 | Materials and Methods

Based on the pulse-tube cryocooler (TeslatronPT from Oxford Instruments Inc.) IntegraAC, recondensing helium magnet system from Oxford (the dry magnet system) with a variable temperature insert (VTI), the instrument operates at the maximum magnetic field of 12 T and the minimum temperature of 1.5 K (shown in Figure 1a,b). To insert the RSTM into the magnet, the transportation utilizes the probe, which is made of the switch box at top for connecting the RSTM with the controller, the flange for sealing the probe and the magnet together, the stainless steel tubes as the main part for decreasing the distance between the center of the magnet to the sample on the RSTM and configuring the twisted pairs inside themselves, the shield



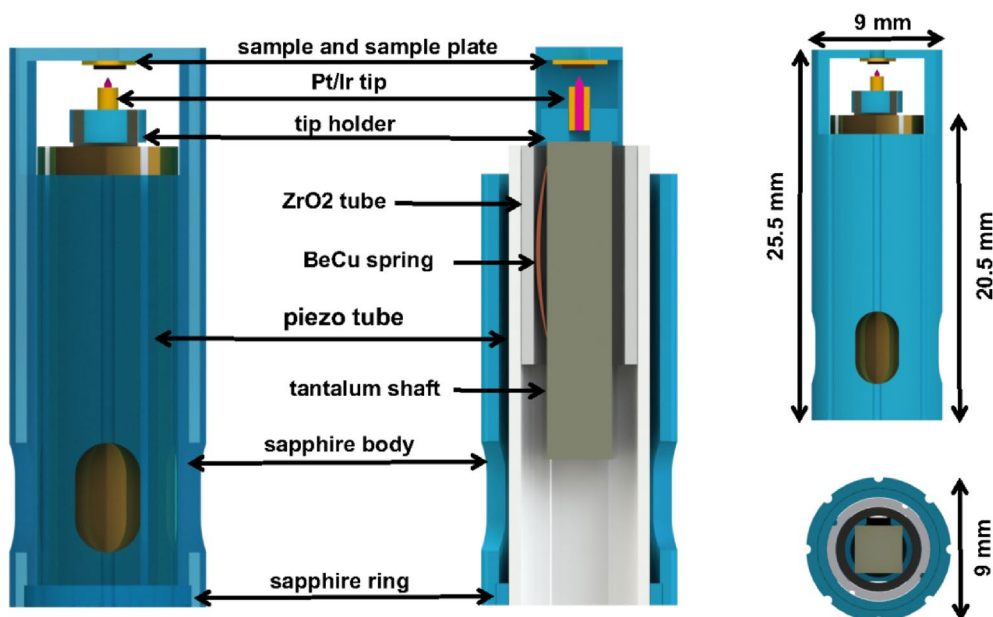
**FIGURE 1** | The overall cutline of the RSTM in the dry magnet and the magnet provides the harsh environment because of the cold head. (a) The display of the dry magnet system utilized in the development. (b) The profile of the RSTM system and the structure of the magnet. (c) The layout of the probe and the key technology is based on the vibration-resistance structure; the spring connected heavy block connecting the scanning head with another small spring.

to block the thermal radiation from the top of the VTI, the copper heat sink for stabilizing the temperature inside the sample chamber, the sapphire ring with pins for connecting the RSTM with the probe, and the spring-connected block for eliminating extreme vibration from the magnet (shown in Figure 1c). To help decreasing the perturbation very effectively, the size of the spring is chosen to be 18 mm for the outer diameter and 1.2 mm for the diameter of the wire, ensuring that the spring can be sturdy enough to hold the heavy block with the eigenfrequency of the oscillation to be around 1 Hz before the probe is inserted into the magnet. Besides, from the simulation based on the weight of the scanning head and the heavy block and the properties of the springs, the simulation has estimated that under the perturbation of around 0.1 N and 10 Hz, the position of the scanning head and the heavy block will remain stable within 20 s, guaranteeing the effectiveness of the vibration damping structure (Figure A1a,b).

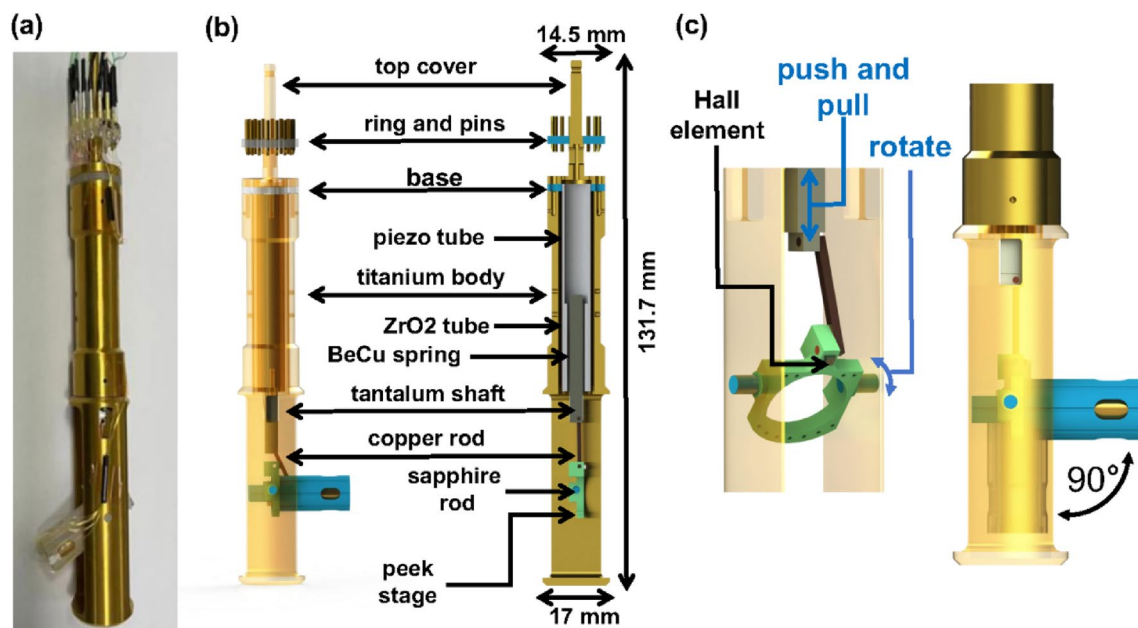
Then, the design of the scanning head and its realization of the rotation are discussed. The following content will focus on the structure at first and the scanning head can be segmented into two units: the scanning unit and the rotating unit. The scanning unit is integrated with several parts shown in Figure 2: with a sapphire ring, which connects the piezo tube and the sapphire body together, at the end, the 20.5 mm length piezo tube whose outer diameter is 6.44 mm and inner diameter is 5.35 mm (four electrodes outside, 0.045 mm thickness, EBL#3 from EBL Products Inc.) consists of the most critical part. The shaft can be driven by the piezo tube because the BeCu spring enforces the tantalum shaft to contact the polished inner surface of the zicornia tube to offer sufficient friction to overcome the gravity while mild enough to generate the relative displacement between the tube and the shaft under the swift altering of the

piezo tube. At the top of the shaft, the sapphire stage insulates the shaft and the Pt/Ir tip. As for the placement of the sample, it is achieved by applying a copper plate, which is utilized to affix the sample with conductive glue, at the top of the sapphire body. The rotating unit is amalgamated with diverse pieces (shown in Figure 3a,b): with the titanium body with nonmagnetic coating (maximum diameter is 17 mm and the minimum diameter is 14.5 mm) outside, the 52 mm piezo tube, whose outer diameter is 7.09 mm and inner diameter is 6.00 mm (four electrodes outside, 0.9 mm thickness, EBL#3 from EBL Products Inc.), fixes a sapphire ring at the end, combining the polished zicornia tube and the BeCu spring together to drive the shaft. Connecting the copper rod, which converts the movement of shaft to the rotation of to peek stage, around the joint at the top, the shaft push and pull the peek stage as Figure 3c to change the orientation of the scanning unit. The critical point for smoothly rotating the peek stage while stabilizing the scanning unit at exact position is to utilize the polished scanning rod as the joint between the stage and the titanium body and fix the stage and the rod with appropriate amount of the cleanable glue to keep balance between movement and stability. The rotation of the scanning unit can range from 0° to 90° and the rotating is continuing without retreating the tip (keep the piezo tube to be relaxing with no voltage). Besides, taking advantage of the piezo tube motor to drive the scanning unit instead of rolling the magnet or the scanning unit by mechanical way, the design can avoid complicated structure and reach high-accuracy adjusting of the angle with appropriate methods to acquire the accurate angle.

Here, the performance of the Hall element to measure the angle when rolling the scanning head will be introduced. The GaAs Hall element (HG-106C, AsahiKASEI MICRODEVICES) will response to the action of the magnetic field with the Hall



**FIGURE 2** | The layout of the scanning unit of the RSTM. The scanning unit is only 25.5 mm in length with 9 mm diameter. The scanning and the approaching are based on the 20.5 mm piezo tube.

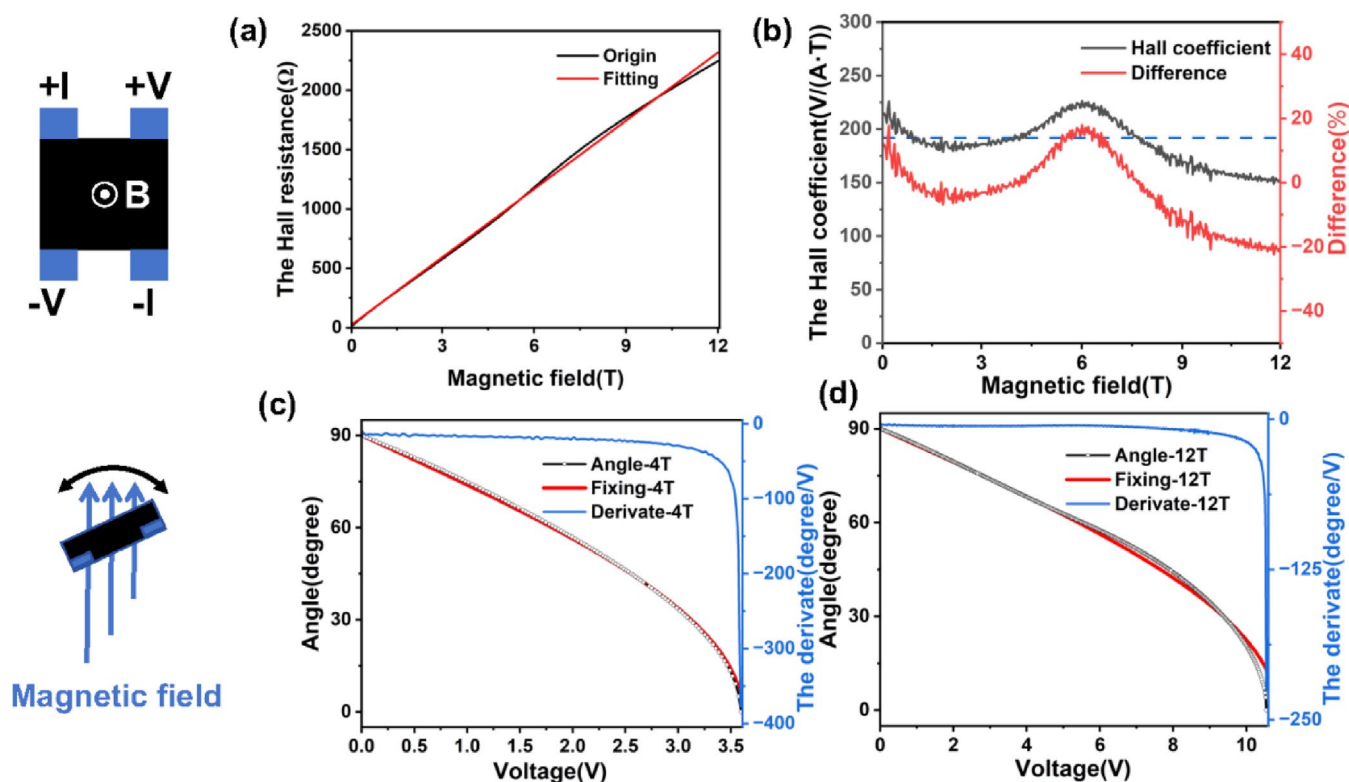


**FIGURE 3** | The layout of the scanning head and the schematic plot of the rotating part. (a) The photo of the scanning head and the scanning head is connected with many soft wires for controlling. (b) The basic structure of the scanning head. (c) The schematic explanation of the rotating with this structure.

voltage when it is activated by the external resources of current. In the experiment, the external excitation voltage and the current set to be 6 V and 4.7 mA for both the room temperature and the low temperature. At the room temperature, the competence of the Hall element has been examined at low vertical field for 0.5 and 1 T, and the voltage increases linearly from 0.86 to 1.72 V indicating the appropriate relationship between voltage and the magnetic field. The magnetic resistance has been examined by the physical property measurement system (PPMS) at 2.1 K up to 12 T to assure the capability for the Hall element at low temperature and high magnetic field, and the

correlation of Hall resistance and the field has also been examined in Figure 4a with the linear fitting. Then, the fluctuation of the influence of the field to the Hall coefficient by extracting the coefficient from the Figure 4a and the wave-like fluctuation of the Hall coefficient in Figure 4b means that the flexible value will cause the error up to 20% compared with the result from the PPMS, indicating an extremely large error for accurate measurement. The evaluation of the resolution of the Hall element is based on the relationship between the Hall voltage and the angle. The resolution at 4 and 12 T in Figure 4c,d is derived by assuming that the horizontal field





**FIGURE 4** | The performance of the hall element to measure the angle based on the measurement of PPMS. (a) The relationship of the hall resistance versus the magnetic field, the black line is the raw data and the red line is the linear fitting. (b) The hall coefficient and the difference between the actual data and the fitting value based on the result of (a); the black line is the hall coefficient and the red line is the difference of the black line minus the fitting slope; (c) the variation of the angle and the derivate of the angle versus voltage at 12 T; (d) the variation of the angle and the derivate of the angle versus voltage at 4 T. The derivate represents the sensitivity of the variation of the angle per unit voltage.

will not affect the output of the Hall element and based on the excitation and the result of the Hall coefficient. By tilting the field from the in-plane orientation to the out of plane orientation, the variation of the angle will be gradually enhanced. The resolution, which means the change of the angle per volt, also follows the same tendency, and it shows the excellent performance of the resolution of the angle especially for the horizontal field, which can stimulate the exotic phenomena especially for the anomalous pairing in superconductivity (Cho et al. 2021). Limited by the voltmeter (the effective resolution is 0.01 V) and the controlling signal, the resolution of the angle of rotating around 4 T at 1.73 K—the condition that the actual rotating experiment is performed—from Figure 4d is 0.11° per 0.01 V near the horizontal position if the closed-loop controlling can be applied. In the following experiment, Figure 4d will be taken as the reference to get the discrete angle for the measurement of spectrum for graphite at 4 T.

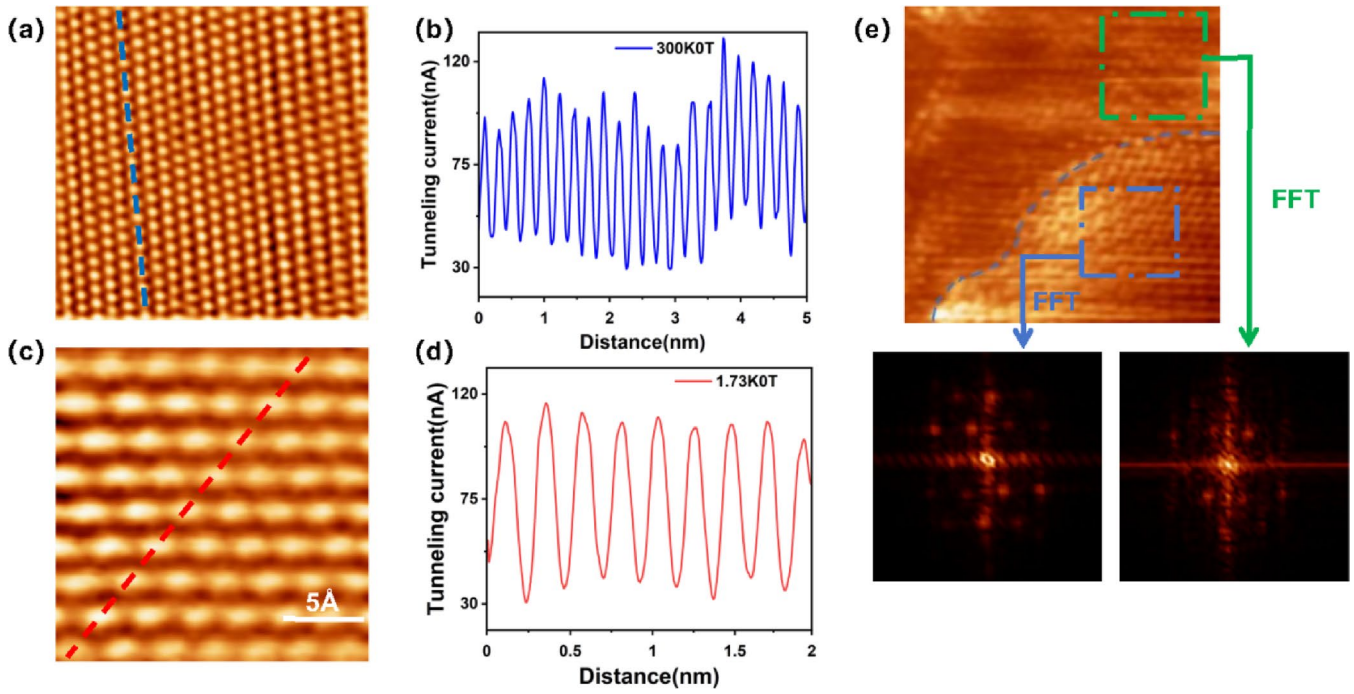
For the sample tests, two samples are applied to examine the functionality of the device for imaging and obtaining spectrums. The silver paint (SPI conductive silver paint 05001-AB) is utilized to stick the sample to the copper plate for good conductance and the steady fixing and stabilize the copper plate to the scanning head, and then the scanning head is inserted into the dry magnet with the probe. The imaging is firstly tested at the room temperature to ensure the successful cleavage of the sample and the excellent resolution from the hand-made Pt/Ir tip. After the retraction of the tip, the temperature goes down by

utilizing the VTI for the low-temperature experiment. The imaging and the spectrum will be taken before rotating the scanning unit at the low temperature. Then the rotation is examined at the low temperature by setting the voltage at z direction of the piezo tube of the scanning unit to be zero after scanning, and then the voltage is applied to the piezo tube of the rotating unit to different orientation. The result is exhibited at different magnetic fields in the next segment.

### 3 | Result and Discussion

The performance of the piezo tube for rotating is examined at first. The experiment is set at the room temperature to verify the rotation from 0° to 90° is applicable. By applying the 60 V voltage to the four outer electrodes and the reference zero voltage, the motor drives the rotation smoothly and can maintain at the exact angle stably. The low temperature result is introduced later at the experiments of samples with the magnetic field. For the behavior of the motor, the critical temperature (463 K) is not investigated here because the application of this motor is at the low temperature and even the sub-Kelvin temperature.

The result of the samples focuses on two different materials: bulk graphite and bulk NbSe<sub>2</sub>. First, the topography of the graphite at both 300 and 1.73 K without field is shown in Figure 5a,c,e. The surface is smooth from the topography and the profiles (Figure 5b,d) when scanning at relatively small

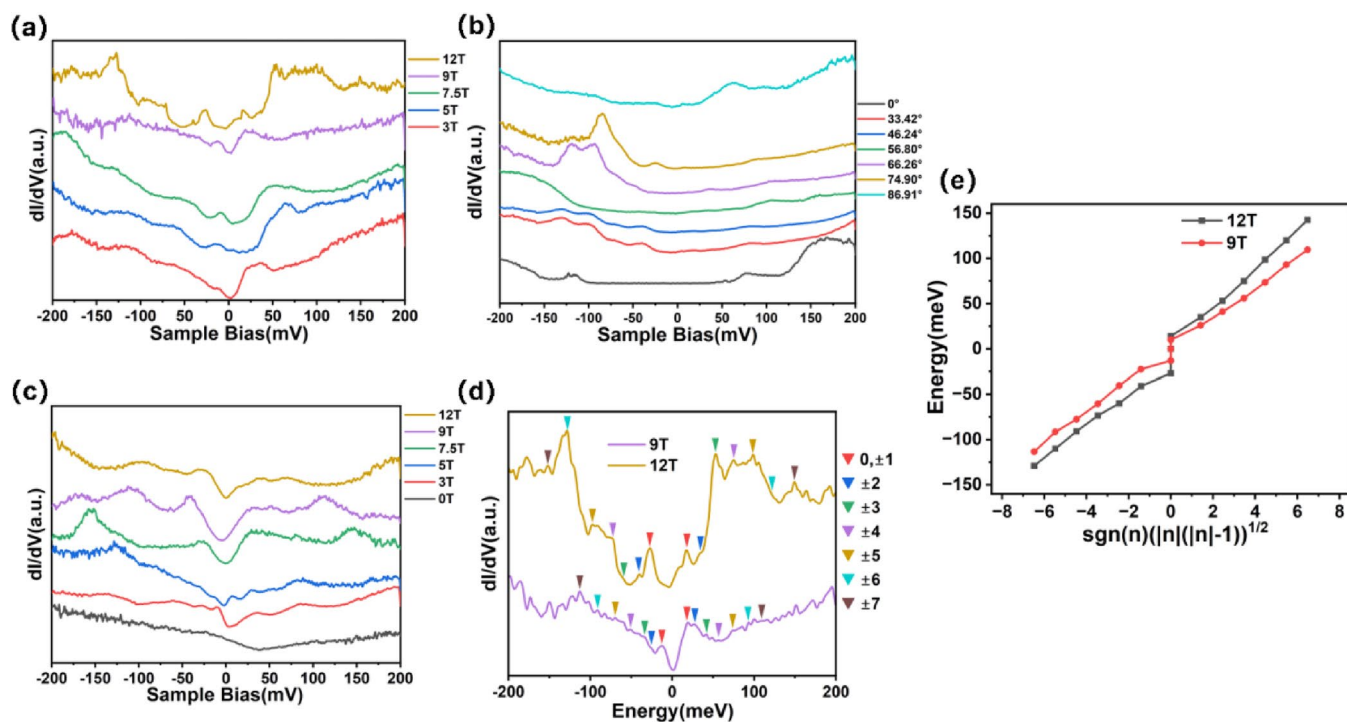


**FIGURE 5** | The measurement of the image of the graphite. (a) The  $5.4 \times 4.7$  nm scanning range of graphite at 300 K, the blue line is the profile line for (b). (b) The profile of the graphite at 300 K. (c) The  $1.5 \times 1.5$  nm of graphite at 1.73 K, the red line is the profile line for (d). (d) The profile of the graphite at 1.73 K. (e) The boundary for two different structures of graphite at 1.73 K for the large scanning range. Blue square contains two  $60^\circ$ -twisted lattice and a superstructure  $\sqrt{3} \times \sqrt{3}$  while the green one only contains single lattice and a superstructure  $\sqrt{3} \times \sqrt{3}$ .

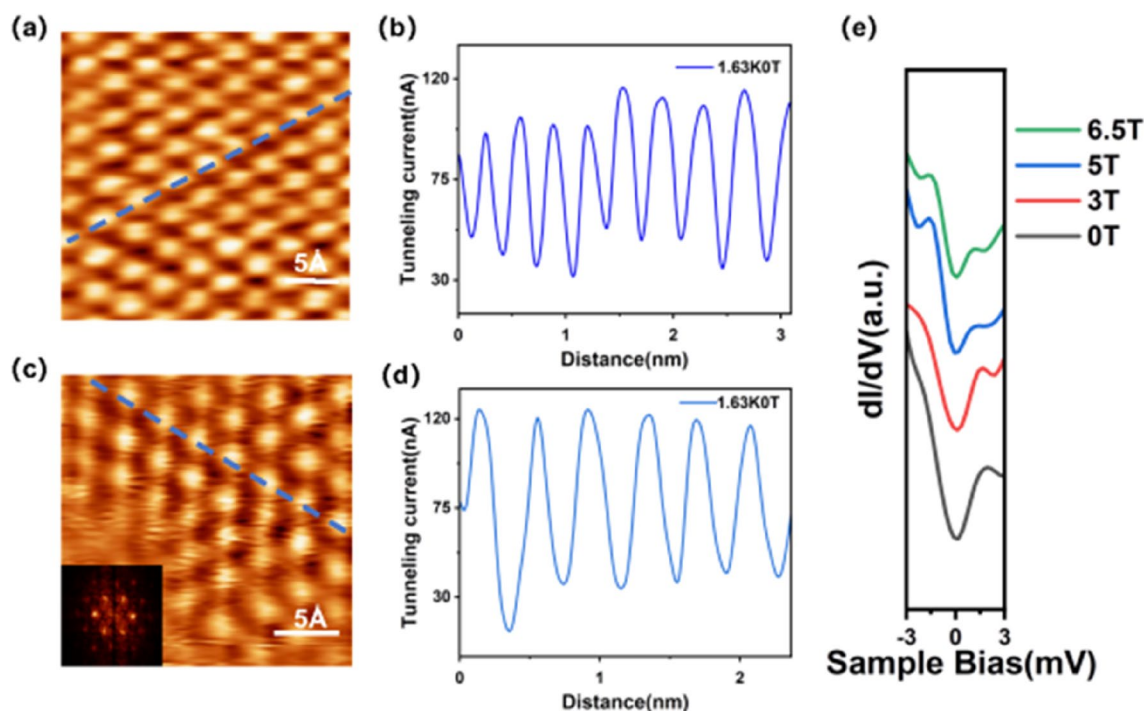
range. However, when turned to another area, the image shows the boundary separates two different topographies: one is composed of a superlattice and two lattices with the twisted angle of  $60^\circ$ , and another is only composed of single lattice and the superstructures (shown in Figure 5e); the scanning unit still has the capacity of recognizing the superstructure, indicating that it can be applied to the researches of the complex surface structure at low temperature. The formation of these different structures can be explained by the interaction of the electrons between layers (Wang et al. 2015). Then, the field is applied up to 12 T for both horizontal and vertical direction, and the rotating field is applied to 4 T at 1.73 K to modulate the spectrum on the smooth surface. The spectrum is obtained at the vertical field from 3 to 12 T at first (shown in Figure 6a), and the fluctuation of the LLs emerges especially when the field rises up to 9 and 12 T. By marking the peaks on the fluctuation and analyzing the data with the relationship of the filling level  $n$  versus energy  $E$  at 9 and 12 T (shown in Figure 6d), the LLs fit the relation of the massive Dirac electrons (Li and Andrei 2007)  $E = \text{sgn}(n)\hbar\omega_C |n| / (|n| - 1)$  ( $n = 0, \pm 1, \pm 2, \pm 3, \dots$ ) where  $\omega_C = \frac{eB}{m_e^*}$  ( $m_e^*$  is the effective mass of Dirac electrons) in Figure 6e. On the contrary, the influence of the in-plane magnetic field affects little to the surface electronic states, and the spectrum remains stationary even the field up to 12 T in Figure 6b. The output of Hall voltage is 0 at that point when rotating the scanning unit at 0.5 T, and the result of Figure 6b proves that the scanning unit is rotated to the in-plane orientation and the good performance of the angle-measuring system. The final step is to measure the different orientation of influence of the magnetic field to the graphite at 4 T. The scanning unit rotates at 4 T, and the spectrum is obtained at

different directions (at  $0^\circ, 33.42^\circ, 46.24^\circ, 56.80^\circ, 66.26^\circ, 74.90^\circ, 86.91^\circ$ ) when referring the Figure 4c. The number of the voltmeter will gradually change when the orientation is varying and will be stable when the driver stops approaching, ensuring that the magnetic field has little effect to the functionality of the driver and will not disturb the position at the certain angle. The observation of the abnormal shifting of the peaks (around  $-100$  and  $-20$  meV) when the field gradually changing its orientation also provides the evidence for the effective rolling of the scanning unit (shown in Figure 6c). As for the accuracy of the rotation of the angle, although the voltmeter can recognize the minimum Hall voltage of 0.01 V, the error brought by the mechanical structure of the joint around the stage still needs to be considered (the hole is 0.05 mm larger than the diameter of the copper rod) and after the geometrical calculation the error is simulated to be around  $\pm 0.24^\circ$  if too extreme perturbation is involved (the wobbly scale of voltmeter to be  $\pm 0.02$  V at tiny-twisted angle at 4 T).

Then the NbSe<sub>2</sub> is tested by the scanning head. Here, only the smooth topography at 1.63 K is imaged at 0 T as shown in Figure 7a,b. Then the scanning range is limited to the smaller area with defects but perfect hexagonal lattice as shown in Figure 7c. Then the rotating of the scanning head to the in-plane orientation is to check the variation of the gap. By applying the field up to 6.5 T, the order parameter of the NbSe<sub>2</sub> can be slightly decreased but will not vanish because of the high upper critical field (up to 10.6 T for homogeneity (Yasuzuka et al. 2020)) in the in-plane direction, keeping the similar tendency as the behavior of the in-plane spectrum of the few-layer NbSe<sub>2</sub> (Dvir et al. 2018). The measurement of the gap of superconductor also



**FIGURE 6** | The rotatable measurement of the spectrum of graphite. (a) The spectrum of out-of-plane magnetic field from 3 to 12 T at 1.73 K. (b) The spectrum of in-plane magnetic field from 0 to 12 T at 1.73 K. (c) The 4 T-rotating spectrum at 1.73 K. There are two peaks (around  $-20$  and  $-100$  meV) shifting with the variation of the magnetic field. (d) The peaks of the Landau levels (LLs) of massive Dirac electrons at 9 and 12 T in (a). (e) The relationship of the LLs in (c) versus  $\sqrt{|n|(|n|-1)}$ . The spectrum is obtained at modulation frequency of 1.31 kHz and amplitude of 3 meV with tunneling current of 2.1 nA.



**FIGURE 7** | The spectrum and the image of NbSe<sub>2</sub>. (a) The  $2.8 \times 2.7$  nm image of the bulk NbSe<sub>2</sub> at 1.63 K, the blue line is the profile line for (b). (b) The profile of the large area of NbSe<sub>2</sub> at 1.63 K. (c) The  $2 \text{ nm} \times 2 \text{ nm}$  image of the bulk NbSe<sub>2</sub> at 1.63 K, the blue line is the profile line for (d). (d) The profile of the large area of NbSe<sub>2</sub> at 1.63 K. (e) The order parameter for NbSe<sub>2</sub> at 1.63 K from 0 to 6.5 T at the in-plane orientation.

gives further confirmation of the successful rotating at the dry magnet at low temperature. For the further improvement, obtaining the spectrum of the NbSe<sub>2</sub> up to 12 T and examining the

influence of the spectrum at higher magnetic field in different direction can be promising especially for the investigation of critical behaviors.



## 4 | Conclusion

This article has introduced the development of a compact and stable RSTM in the dry magnet (12 T maximum field and 1.5 K minimum temperature), and it can reach the accuracy of the controlling of angle to  $0.11^\circ$  per 0.01 V if the closed-loop controlling can be applied based on the measurement at 4 T with the activation to Hall element at 6 V and 4.7 mA and has the rotating range from  $0^\circ$  to  $90^\circ$ . The accuracy is determined by the evaluation of the Hall coefficient and the resistance of the Hall element by PPMS and the steady scale of the voltmeter (0.01 V available). The application of the rotating unit can also be applied to other types of scanning probe microscopes including the MFM and the spin-polarized STM. With the compact structure and the stable vibration-resistant probe, the scanning head can be applied to various narrow spaces promisingly inside the harsh magnet including the water-cool magnet and the hybrid magnet in the future, and it can also contribute to the researches of STM in the cryogenic-free system. The multiangle measurement of the spectrum of the graphite and the NbSe<sub>2</sub> and the visualization of the topography clearly exhibit the powerful functionality of field-rotatable investigation, and we have observed the obvious changing of the LLs with the modulation of the orientation of the magnetic field and the stability of the gap under the low magnetic field compared with the critical in-plane field of bulk NbSe<sub>2</sub>. The most significant capability for the device is to accurately adjust the in-plane magnetic field, which is very critical to the uncovering of the exotic phenomena like the FFLO state (Cho et al. 2021; Dvir et al. 2018; Kumagai et al. 2006; Uji et al. 2006) and the in-plane anomalous Hall effect (Zhou et al. 2022), providing a new dimension to investigate the quantum materials by STM.

## Author Contributions

**Junwei Liu:** writing – original draft, visualization, formal analysis, methodology, investigation, conceptualization, data curation, software. **Zihao Li:** investigation, formal analysis, data curation, software. **Dan Wu:** investigation, formal analysis, data curation. **Shaofeng Zheng:** investigation, software, data curation. **Yue Gao:** investigation, formal analysis. **Shiwei Yang:** investigation, formal analysis. **Jihao Wang:** conceptualization, resources. **Jing Zhang:** investigation, resources. **Qiyuan Feng:** investigation, resources. **Yubin Hou:** conceptualization, resources, project administration. **Wenjie Meng:** conceptualization, validation, project administration, resources, supervision, writing – review and editing, software, methodology, visualization, formal analysis. **Yalin Lu:** resources, project administration, supervision. **Qingyou Lu:** supervision, project administration, resources, conceptualization, methodology, writing – review and editing, validation.

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## Conflicts of Interest

The authors declare no conflicts of interest.

## Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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## Appendix A

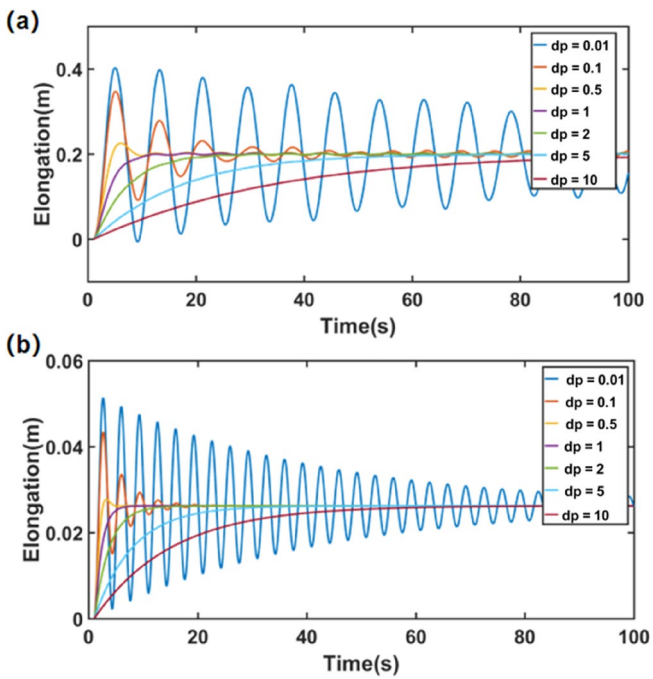
When estimating the capability of eliminating the perturbation by the secondary counterweight, we have utilized the simple model below:

$$m_1 \frac{\partial^2 x_1}{\partial t^2} = m_1 g + k_2 x_2 - k_1 x_1 - \beta_1 \frac{\partial x_1}{\partial t} + A_1 \sin(\omega_0 t) \quad (1)$$

$$m_2 \frac{\partial^2 x_2}{\partial t^2} = m_2 g - k_2 x_2 - \beta_2 \frac{\partial x_2}{\partial t} + A_2 \sin(\omega_0 t) \quad (2)$$

where  $m_1$  and  $m_2$  are the mass of the heavy block and the scanning head;  $x_1$  and  $x_2$  are the elongation of the large spring and the small spring;  $k_1$

and  $k_2$  are the spring constant of the large spring and the small spring;  $\beta_1$  and  $\beta_2$  are the damping coefficient of the large spring and the small spring;  $A_1$  and  $A_2$  are the amplitude of the perturbation of the large and small spring, and  $w_0$  and  $\omega_0$  are the frequency of the perturbation. We have estimated that mass of the heavy block to be 0.265 kg and the mass of the scanning head to be 0.049 kg, and the amplitude and the frequency of the perturbation are same as  $A_1 = A_2 = 0.1N$  and  $w_0 = \omega_0 = 10Hz$  and the spring constant is  $k_1 = k_2 = 17.5N/m$ . We can realize that the perturbation of both situations can reduce to zero swiftly.



**FIGURE A1** | The performance of eliminating the external perturbation. (a) The simulation of the larger spring for reducing the mechanical vibration and the suitable damping can result in the perfect reduction of the noise; (b) the simulation of the smaller spring for reducing the mechanical vibration. The attenuation of noise still performs well with the external perturbation.